

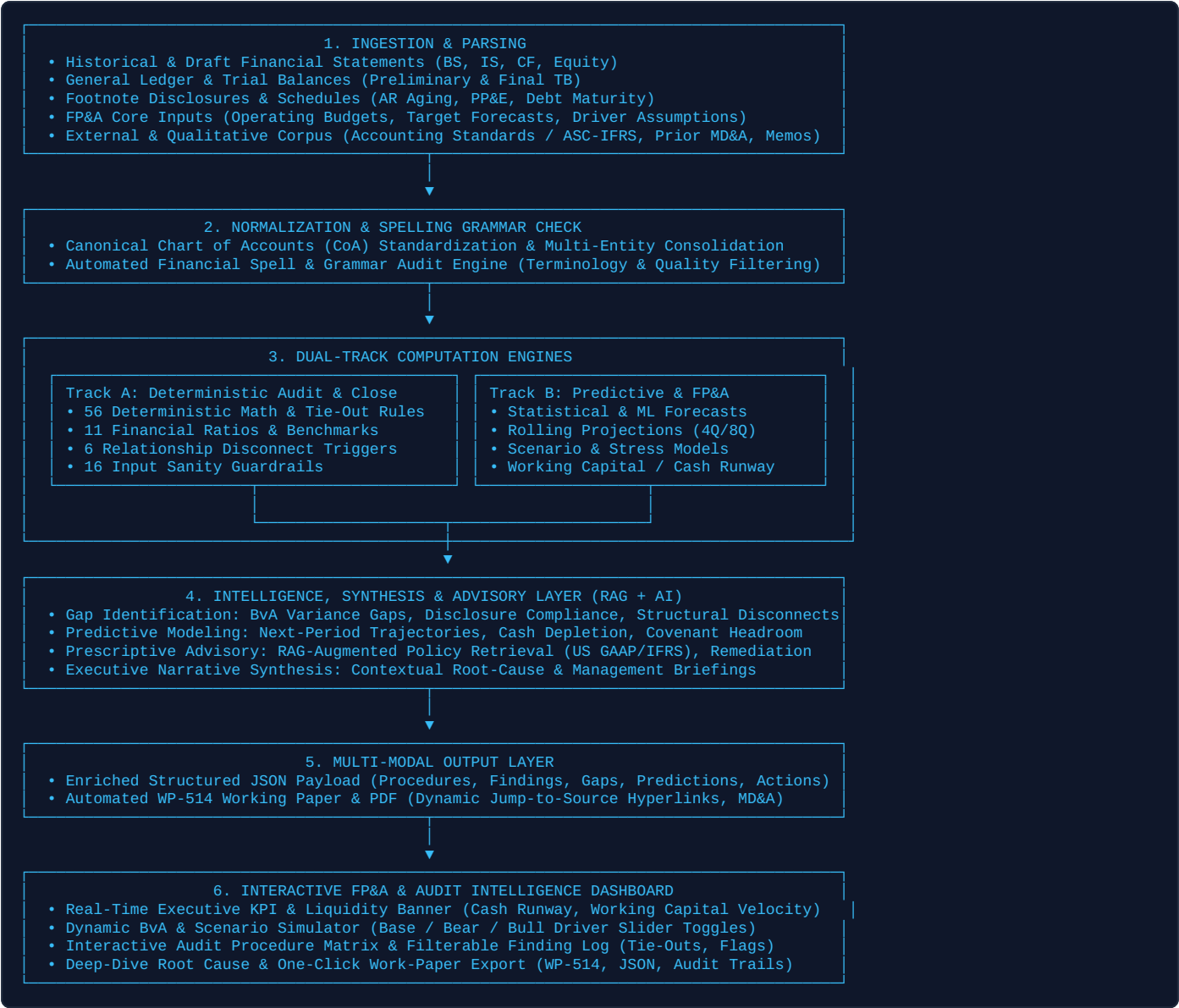
Enterprise Architecture Specification

Complete Technical Blueprint: End-to-End FP&A, Audit Assurance, Intelligence Suite & Interactive Dashboard

Domain: Banking & Financial Services	Role: Technical Lead & FP&A Expert	Framework: US GAAP / IFRS / Bank Directives	Document: ARCH-SPEC-WP514-v3
---	---	--	-------------------------------------

This specification provides an exhaustive, production-grade architectural and implementation blueprint for an end-to-end **Financial Planning & Analytics (FP&A)** and **Financial Statement Review & Audit Assurance Engine**. It integrates data normalization with automated spell/grammar verification, deterministic closed-loop tie-out controls, predictive driver modeling, semantic RAG policy governance, multi-modal outputs, and an interactive executive command dashboard.

1. SYSTEM ARCHITECTURE OVERVIEW



2. DETAILED LAYER-BY-LAYER ENGINEERING BREAKDOWN

Layer 1: Ingestion & Parsing Layer

Functional Responsibility: Multi-modal ingestion of heterogeneous structured, semi-structured, and unstructured financial records across reporting periods.

- **Historical & Draft Statements:** Ingestion of Balance Sheet (BS), Income Statement (IS), Cash Flow (CF), and Statement of Stockholders' Equity across prior audited baselines and current draft closes.
- **Trial Balances:** General ledger extract ingestion covering Preliminary Trial Balance (pre-closing) and Final Trial Balance (post-adjusting journal entries).
- **Footnote Schedule Extractors:** Granular tabular extraction for Accounts Receivable Aging (Current, 1–30, 31–60, 61–90, >90 days), PP&E Roll-Forward (Gross, Accumulated Depreciation, Additions, Retirements, Net PP&E), and Debt Maturity Schedules (Year 1 to 5 + Thereafter).
- **Forward-Looking FP&A Inputs:** Ingestion of Annual Operating Budgets (AOB), 4Q/8Q dynamic rolling forecasts, and operational volume/headcount driver assumptions.
- **Qualitative Regulatory Corpus:** Ingestion of US GAAP (ASC Topics), IFRS frameworks, bank credit risk directives, prior period MD&A commentary, and board oversight memos.

Layer 2: Normalization & Spelling Grammar Check Layer

Functional Responsibility: Harmonizing disparate Chart of Accounts naming structures and auditing textual quality across financial reports.

- **Canonical Chart of Accounts (CoA) Standardization:** Normalizes inconsistent client nomenclature (e.g., "*Top-line Sales*", "*Gross Revenues*", "*Operating Turnover*") into a unified canonical taxonomy (**Revenue**) conforming to standardized reporting nodes across multi-entity consolidations.
- **Automated Financial Spelling & Grammar Audit:** Executes domain-specific linguistic verification on ingested disclosures, footnotes, and management notes. Detects typographic errors, non-standard financial abbreviations, syntax flaws, and inconsistent signage conventions prior to downstream computation.

Layer 3: Dual-Track Computation Engines

Functional Responsibility: Concurrent execution of deterministic audit verification and dynamic forward-looking FP&A forecasting.

Track A — Deterministic Audit & Close Engine (Zero Hallucination):

- **56 Deterministic Rules:** Complete execution suite comprising 11 Mathematical footings, 4 Cross-statement tie-outs, 5 Prior-year comparatives, 8 Footnote schedule ties, 4 Analytics metrics, 11 Financial ratios, 1 Materiality flagging filter, and 6 Relationship disconnect checks.
- **16 Input Sanity Guardrails:** IAS/IFRS sanity limits covering operational cash buffers, retained earnings continuity, CapEx/OpEx scaling, and tax rate consistency.

Track B — Predictive & Driver-Based FP&A Engine:

- **Budget vs. Actual (BvA) Engine:** Computes absolute and percentage plan deviations ($\Delta = \text{Actual} - \text{Budget}$, $\% \Delta = \frac{\Delta}{\text{Budget}} \times 100\%$).
- **Rolling Projections (4Q/8Q):** Dynamic multi-period forecasting integrating regression baselines with operational driver assumptions.
- **Liquidity & Runway Velocity:** Computes net monthly cash burn and runway exhaustion timeline:

$$\text{Cash Runway (Months)} = \frac{\text{Total Liquid Cash \& Equivalents}}{\text{Monthly Net Negative Operating Cash Flow}}$$
- **Scenario Sensitivity Modeling:** Simulates Base, Bear, and Bull macroeconomic shocks on debt-service capacity.

Layer 4: Intelligence, Synthesis & Advisory Layer (RAG + AI)

Functional Responsibility: Augmenting deterministic audit exceptions with contextual root-cause synthesis, compliance standard citations, and actionable recommendations.

- **Gap Identification Engine:** Detects operational BvA shortfalls, disclosure compliance omissions (e.g., missing contingent liabilities), and structural margin disconnects (e.g., inventory expansion outpacing COGS growth).
- **Predictive Trajectory Engine:** Produces forward-looking confidence intervals for top-line revenue, gross margins, and debt covenant clearance headroom.
- **Prescriptive Advisory Engine (RAG-Augmented):** Retrieves exact accounting clauses from ASC/IFRS standards (e.g., ASC 326 CECL provisioning, ASC 842 lease capitalization) to formulate prioritized corrective remediation steps.
- **Executive Narrative Synthesis:** Compiles executive-grade MD&A summaries contextualizing financial results and audit findings for executive leadership.

Layer 5: Multi-Modal Output Layer

Functional Responsibility: Delivering machine-readable API payloads and publication-ready audit deliverables.

- **Enriched JSON Payload:** Machine-readable export conforming to `structured_output_template.json` enriched with `gap_analysis`, `predictions`, `recommendations`, and `source_citations`.
- **Interactive WP-514 PDF Working Paper:** Compiles executive MD&A summaries, ratio dashboards, and clickable deep jump-to-source hyperlinks navigating directly to source cell coordinates or statement pages.

Layer 6: Interactive FP&A & Audit Intelligence Dashboard

Functional Responsibility: Providing finance executives and audit teams with a centralized, real-time command center for monitoring, stress-testing, and root-cause exploration.

- **Executive KPI & Liquidity Banner:** Real-time display of critical financial vitals: Quick & Current Ratios, Cash Runway (months remaining), Operating Margin, and Cash Conversion Cycle (CCC) with dynamic status color badges (Green / Amber / Red).
- **Interactive BvA & Variance Heatmap:** Drillable visual hierarchy displaying Budget vs. Actual variances across departments, business units, and statement line items with configurable variance thresholds.
- **What-If Scenario & Stress-Test Simulator:** Interactive slider controls allowing FP&A leads to adjust operational drivers (e.g., $\pm 10\%$ sales volume, price increases, interest rate shifts) and immediately visualize recomputed forecast trajectories and cash burn curves.
- **Audit Verification & Flag Console:** Searchable, sortable procedure matrix showing the status of all 56 rules and 16 guardrails, complete with confidence scores, severity badges (Low/Medium/High/Critical), and instant review/waiver status toggles.
- **Deep-Dive Root Cause Explorer & Traceability View:** Split-screen view allowing users clicking any flagged finding or gap to instantly view extracted source tables, accounting rule references (ASC/IFRS), and recommended resolution steps side-by-side.
- **One-Click Export & Integration Hub:** Instant export of generated WP-514 working papers (PDF), structured JSON payloads, and formatted Excel tie-out schedules, with webhooks connecting directly to ERP/GL platforms.

3. PROJECT IMPLEMENTATION ROADMAP & ACTION PLAN

Phase	Module / Workstream	Key Technical Tasks & Deliverables	Domain Standard
PHASE 1	Ingestion & Parsing	Build robust extractors for BS, IS, CF, Equity, TBs, Budgets, and Footnotes (AR Aging, PP&E, Debt Maturity).	Data Pipeline / ETL
PHASE 2	Normalization & Spell-Check	Implement Canonical CoA dictionary standardization and domain-specific financial spell/grammar validation engine.	Data Quality & Governance
PHASE 3	Deterministic Rule Suite	Consolidate 56 rules and 16 input guardrails into a pure Python execution class with zero AI hallucination.	SOX / Audit Controls
PHASE 4	FP&A Analytics Engine	Implement BvA variance engine, 4Q/8Q rolling projection algorithms, and dynamic cash runway calculators.	Corporate FP&A / Treasury
PHASE 5	RAG & Advisory Layer	Index US GAAP/IFRS standards; build structured prompt chains for Gaps, Predictions, and Recommendations.	ASC/IFRS Compliance
PHASE 6	JSON & Interactive PDF	Extend JSON schema; generate WP-514 PDF with embedded clickable jump-to-source link annotations.	WP-514 Working Papers
PHASE 7	Executive Dashboard	Build interactive React/Next.js or Streamlit web dashboard with KPI cards, scenario simulators, and audit consoles.	Executive Intelligence